AAAI/ACM AIES 2026 Co-Located Workshop

PROCEEDINGS OF CONVERGING PERSPECTIVES ON HEALTH AI (CPHAI)

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Introduction

Over the last decade, artificial intelligence (AI) systems have been increasingly deployed in the healthcare domain and have become commonplace within conversations around health. Our focus is on AI deployed in health and care settings - systems that enter diagnosis, treatment, and the work of healthcare providers, while remaining attentive to the wider conditions of health in which these systems operate. As AI systems rapidly reshape the healthcare environment, meaningful impact, not merely good performance, becomes the critical concern. This has led to a growth in research pertaining to, among others, explainable and trustworthy AI. Nevertheless, many existing AI systems still fail to deliver on their promises in healthcare workflows and other real-world use cases, often because legal, ethical, and epistemic challenges have not been integrated into their development and deployment.

Furthermore, AIES provides a groundwork for sharing research across computer and social sciences, law and policy, ethics, and philosophy. Within each of these domains, AI systems for health are developed and evaluated under different definitions of accountability, ethics, trustworthiness, safety, and compliance. The result is that each domain remains siloed from the others, with little articulation across the health AI field. Yet this articulation is essential for the development and evaluation of real-world AI systems. We count ourselves within this condition rather than outside it: as organizers drawn from computer science, social science, law, ethics, and philosophy, we recognize that the meanings we attach to terms such as accountability, trust, safety, and compliance are themselves shaped by the disciplines that trained us. These are, in a precise sense, essentially contested concepts (Gallie, 1956), and the divergence among our usages is a structural feature of cross-disciplinary work rather than a failure of rigour. We do not assume that a single shared definition can or should be imposed. Instead, we offer this workshop as a trading zone (Galison, 1997) in which differing disciplinary understandings can be surfaced, held side by side, and articulated. What we seek is the convergence of perspectives, understood as an achievement to be worked toward, not a premise to be assumed.

To this end, the Converging Perspectives on Health AI Workshop aims to provide a space for community building centered on the question: How can we develop healthcare AI systems that are clinically useful, technically robust, socially acceptable, and responsibly deployed? Or, put simply, how should we be using or not using AI in real-world clinical settings?

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Table of Contents

INTRODUCTION 1

Acknowledgements 2

Organizing Committee 2

TARGETING, TIMING, AND DURATION: SIMULATING THE LONGITUDINAL EFFECTS OF ACCESS-IMPROVING INTERVENTIONS IN HEALTHCARE AI 4

PRESERVING FUNCTION, QUALITY AND SAFETY IN MEDICAL PROCESS AUTOMATION 9

CONTEXT-AWARE HIERARCHICAL BAYESIAN MODELING OF IVF LABORATORY ENVIRONMENTAL CONDITIONS 14

BEYOND PERSONALIZATION: GOVERNING THE CARE-TO-LEARNING BOUNDARY IN EMBODIED AI 19

MORE THAN BENCHMARKS: MACHINE LEARNING COMPETITIONS AS SOCIOTECHNICAL SYSTEMS 23

WHAT ACCURACY CANNOT CERTIFY: A FAILURE-AWARE EVALUATION FRAMEWORK FOR PATIENT-FACING LLM TRIAGE 28

Targeting, Timing, and Duration: Simulating the Longitudinal Effects of Access-Improving Interventions in Healthcare AI

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Abstract

Healthcare AI systems are embedded in social, institutional, and infrastructural contexts that influence how algorithmic decisions affect patients and how their consequences unfold over time. Yet fairness in health AI is often assessed using metrics computed at a single point in time, potentially overlooking how structural conditions and repeated decisions interact to shape patient trajectories and long-term outcomes. We use longitudinal microsimulation to examine these dynamics and to interrogate the value-laden assumptions embedded in healthcare AI design. Building on our previously developed microsimulation framework, we vary the reach, timing, and duration of interventions designed to improve access to care under five algorithmic allocation policies, each encoding a distinct ethical stance on how benefits should be distributed. Across simulated patient trajectories, the same nominal improvement in access produced different cumulative health consequences depending on when it began and how long it was sustained, and interventions that differed in who they reached produced different reductions in long-term disparity. These findings illustrate how simulation can make otherwise difficult-to-observe temporal and structural dependencies visible and provide a basis for interdisciplinary ethical inquiry. We position longitudinal microsimulation as a complementary tool for evaluating fairness in AI-driven healthcare resource allocation, alongside empirical evaluation and real-world monitoring, by helping identify how disparities may evolve under different policy and intervention choices before deployment.

Code — https://github.com/kwade4/fairness-over-time

Background and Related Work

As healthcare AI is increasingly used to support clinical and resource allocation decisions, understanding its effects requires considering the broader contexts in which these decisions are made. Healthcare decisions do not occur in isolation, and their effects depend on healthcare infrastructure, resource constraints, human decision-makers, and the circumstances of the individuals affected by them. Access to healthcare is often characterized in terms of approachability, acceptability, availability and accommodation, affordability, and appropriateness of care (Levesque, Harris, and Russell 2013). These dimensions can affect whether an individual is able to benefit from an otherwise equitable allocation. Understanding the consequences of healthcare AI therefore requires considering not only the algorithmic decision itself, but also the broader sociotechnical system through which that decision is enacted.

Healthcare is also inherently sequential. Care is delivered through a series of decisions about whether to intervene, which treatment to use, when to adjust therapy, and how closely to monitor progress. These decisions respond to a patient's current condition, treatment history, and changing risks. This is crucial in chronic disease management, where patients interact with the healthcare system repeatedly over months or years, with each encounter influencing subsequent options and outcomes (Jayasooriya et al. 2023; Cross, Choma, and Onofrey 2024). This sequential structure impacts how fairness in health AI should be evaluated.

Fairness is often assessed in terms of how resources or decisions are distributed across individuals or groups, using metrics such as equalized odds, demographic parity, or ensuring error rates are equal across population subgroups (Hardt, Price, and Srebro 2016; Barocas, Hardt, and Narayanan 2023). Audits based on these metrics have uncovered disparities in healthcare AI systems across domains including skin cancer screening (Daneshjou et al. 2022), cardiac care (Mihan, Pandey, and Van Spall 2024), and cost-based risk scoring (Obermeyer et al. 2019). Such analyses evaluate the algorithm's decision at a single point in time, rather than following how that decision plays out over the course of a patient's trajectory. Moreover, receiving an allocation does not necessarily result in receiving care. Access barriers may influence whether an individual can act on or benefit from a resource allocation decision. Since healthcare involves repeated decisions over time, gaps or delays in access can influence future treatment options and long-term health outcomes (Jayasooriya et al. 2023; Cross, Choma, and Onofrey 2024; Naderalvojoud et al. 2025).

Prior work has shown that fairness evaluated at a single decision point may not reflect its longer-term consequences, because decisions can alter individuals' future states and opportunities (Liu et al. 2018). Others further demonstrated how simulation can be used to examine these dynamics and reveal long-term effects that may not be apparent from static evaluations (D'Amour et al. 2020; Alamdari et al. 2024).

Our prior work (Wade, Huang, and Lizotte 2026) examined how different normative fairness objectives shape re-

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source allocation in sequential healthcare decision-making, using quality-adjusted life years (QALYs), a common health economic outcome measure combining survival time and health-related quality of life into a single index (Drummond et al. 2015) as the primary outcome. We found that Egalitarian and Sufficientarian policies, somewhat counterintuitively, produced larger disparities between the best-off and worst-off individuals in QALYs and care received than Utilitarian and Prioritarian policies. This was likely driven by a structural access mechanism; individuals allocated care equally did not necessarily receive it equally, since access to care varied independently of the allocation rule itself. Consequently, allocation policies that appear equitable at the point of decision do not necessarily produce equitable outcomes over time when individuals face different structural barriers to care.

We extend our longitudinal microsimulation to examine how the reach, timing, and duration of access-improving interventions affect long-term health outcomes and disparities. We compare targeted and population-wide interventions introduced early or late and sustained temporarily or permanently. Our results illustrate how simulation can complement conventional fairness evaluation of health AI by making temporal and structural dependencies explicit before deployment.

Methods

Longitudinal Healthcare Simulation

We use the simulation environment and five allocation policies from our prior work (Wade, Huang, and Lizotte 2026) unchanged, and summarize them here for completeness. The new contribution of this paper is the intervention layer described in the next subsection. The simulation represents a population over a 50-year horizon using a Healthy-Sick-Sicker-Dead model (Siebert et al. 2012; Iskandar 2018; Alarid-Escudero et al. 2023) with a continuous latent health representation. Individuals are assigned to five deprivation quintiles, Q0 through Q4 (most to least deprived), and vary in baseline health, care access, and treatment effectiveness. Each month, a fixed healthcare budget is allocated according to a specified policy, after which individuals receive care based on their effective access. Health state transitions, QALYs, and costs are monitored throughout the simulation.

We evaluate five allocation policies, each corresponding to a distinct principle of distributive justice in healthcare ethics. Utilitarian allocation directs resources to whoever has the greatest expected health gain, resembling triage protocols that prioritize patients most likely to benefit under scarce resources. Egalitarian allocation distributes resources equally regardless of need, resembling a strict per-capita distribution rule. Sufficientarian allocation prioritizes bringing individuals below a minimum health threshold up to that floor, resembling public health targets framed around a minimum acceptable standard of care. Prioritarian allocation weights resources toward the worst-off across the full distribution, resembling need-adjusted funding formulas that direct greater resources toward more disadvantaged populations. Needs-Based allocation distributes by current health deficit alone, resembling standard clinical triage by severity.

Access Interventions

We extend the existing simulation by introducing access-improving interventions and evaluating how their targeting, timing, and duration affect longitudinal outcomes. Interventions modify effective care access without permanently changing individuals' underlying characteristics, allowing comparison against a no-intervention baseline under the same policy and inequality structure.

In this study, we investigate three intervention configurations. The targeted intervention represents individual-level support, such as transportation assistance, patient navigation or scheduling support (Haroen et al. 2025; Rangachari, Al Arkoubi, and Shindi 2025; Chen 2026), and increases care access by 0.15 for 70% of individuals in the most deprived quintile (Q0). The systemic intervention increases effective access by 0.05 across the population, representing system-level changes such as expanded clinic hours or additional appointment capacity (Haroen et al. 2025; Chen 2026; Hayre et al. 2026). The combined intervention applies both additive increases simultaneously. Intervention magnitudes are illustrative rather than empirically calibrated. They were selected relative to the baseline access gradient in the simulation (mean access of 0.30 in Q0 and 0.69 in Q4). The targeted intervention provides a larger increase to a smaller group, closing approximately one-quarter of the baseline Q0–Q4 access gap on average, while the systemic intervention provides a smaller increase across the population, although it adds more total access because it reaches everyone.

We compare five timing and duration conditions. Interventions begin either early (year 1) or late (year 25) and last either 5 years (short) or 15 years (long). Permanent interventions begin at year 0 and persist through the 50-year horizon. For the combined intervention, both components follow the same timing and duration schedule. We evaluate each condition under all five allocation policies over a 50-year horizon using a simulated population of 10,000 individuals. We compare cumulative QALYs and the Q4–Q0 QALY gap against a no-intervention baseline to assess aggregate health gains and changes in disparities. We repeated all experiments across three random seeds to assess stability. The main qualitative findings were consistent across seeds. Tables and figures report results from a single seed.

Results

We examined how the timing and duration of targeted access-improving interventions affected longitudinal outcomes, then compared permanent targeted, systemic, and combined interventions under five allocation policies (Table 1). Across intervention types, longer interventions generally produced greater cumulative health benefits than shorter interventions with the same nominal access improvement. However, the timing of the intervention also had a substantial effect, particularly for targeted interventions. Figure 1 shows this pattern under Prioritarian allocation for the most-deprived (Q0) and least-deprived (Q4) quintiles. When the targeted intervention began in year 1 (Panel A), the short 5-year intervention produced a temporary gain in Q0 that decreased after access returned to baseline, while the long 15-year intervention pro-

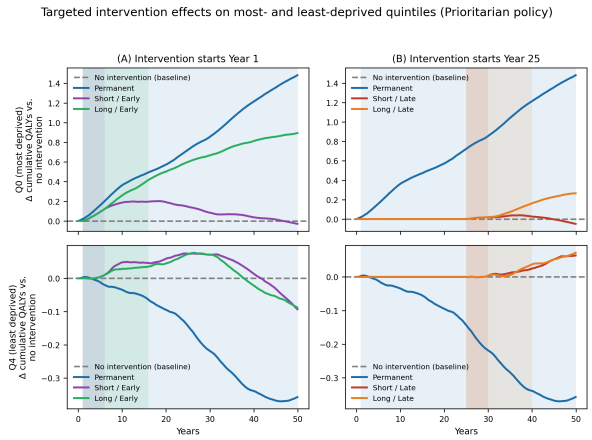


Figure 1: Change in cumulative QALYs relative to no intervention for Q0 (most deprived) and Q4 (least deprived) under a targeted intervention, beginning in (A) year 1 and (B) year 25. Shading indicates the active intervention window.

duced a larger and more persistent gain thereafter. When the same interventions began in year 25 (Panel B), neither condition produced a meaningful cumulative benefit by year 50. Thus, interventions of the same duration can have different long-term effects depending on when access improves.

Long/early conditions produced the largest reductions in the Q4–Q0 gap for the targeted intervention under every policy, while the smallest reduction under each policy occurred with a short intervention (early or late) and was sometimes negative (Table 1). The ordering of intermediate conditions varied somewhat across policies, but the advantage of earlier and more sustained intervention was broadly consistent.

The effects of the intervention also varied depending on how the access improvement was distributed, independent of timing. Table 2 compares the targeted and systemic interventions under permanent implementation. The systemic intervention, which applied a uniform access increase across the population, produced a larger gain in aggregate QALYs than the targeted intervention under four of the five policies (e.g., 3.7% vs. 2.8% under Egalitarian and 1.7% vs. 0.3% under Prioritarian); under Sufficientarian allocation the two were comparable (2.1% vs. 2.2%). However, the targeted intervention produced a larger reduction in the Q4–Q0 gap under every policy (e.g., 19.7% vs. 9.0% under Prioritarian and 28.4% vs. 19.5% under Egalitarian).

The combined intervention did not consistently improve disparity reduction beyond the targeted intervention alone. Under Utilitarian allocation, it produced a larger aggregate QALY gain (1.7% vs. 0.2%) but a smaller reduction in the Q4–Q0 gap (20.8% vs. 23.4%). Because the gap reflects the difference between quintile outcomes, greater aggregate gains need not translate into a larger reduction in disparity. The gap narrows less when additional health gains do not disproportionately benefit Q0 relative to Q4. Under Sufficientarian, Prioritarian, and Needs-Based allocation, the combined intervention instead produced a larger gap reduction than the targeted intervention alone. Under Egalitarian

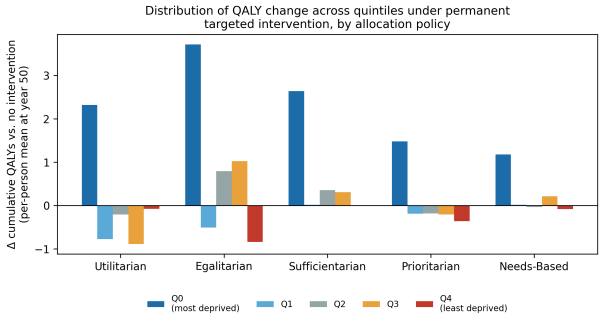


Figure 2: Change in cumulative QALYs at year 50, relative to no intervention, for all five deprivation quintiles under a permanent targeted intervention, for each allocation policy.

Policy	Long/Early	Short/Early	Long/Late	Short/Late
Utilitarian	8.6%	0.5%	6.1%	4.0%
Egalitarian	19.0%	10.2%	1.9%	1.0%
Sufficientarian	11.5%	−3.4%	1.5%	1.0%
Prioritarian	10.5%	0.7%	2.1%	−1.2%
Needs-Based	4.5%	−4.1%	1.3%	−3.3%

Table 1: Percent reduction in Q4–Q0 QALY gap relative to the no-intervention baseline, under the targeted intervention, across allocation policies and intervention timing/duration.

allocation, the gap reductions were nearly identical (28.0% vs. 28.4%). These results suggest that the effect of adding a systemic component on disparity depends on how access improvements translate into care received and health gains across quintiles under each allocation policy.

Since care is allocated from a fixed monthly budget, gains for one quintile can also affect outcomes for other quintiles. Figure 2 shows the change in cumulative QALYs for all five quintiles under a permanent targeted intervention across allocation policies. Q0 gained under every policy, while the remaining quintiles showed more variable changes. Under Utilitarian allocation, for example, Q1 lost 0.78 QALYs compared with a 0.08 QALY loss for Q4. Under Egalitarian allocation, Q1 and Q4 declined by 0.51 and 0.84 QALYs, respectively, while Q2 and Q3 gained. Under Sufficientarian allocation, changes outside Q0 were smaller, with no quintile changing by more than 0.36 QALYs in either direction. These differences illustrate how reallocating access under a fixed-budget assumption can produce policy-dependent changes across the population, rather than uniformly benefiting or disadvantaging particular groups.

Discussion

These findings highlight an important consideration for evaluating AI-enabled healthcare interventions: the effects of an algorithm depend on the temporal and structural context in which its decisions are implemented. Fairness evaluations often focus on how decisions or resources are distributed across groups at a given point in time. For interventions whose effects unfold through repeated healthcare interac-

	Targeted		Systemic		Combined	
Policy	ΔQALY	Gap red.	ΔQALY	Gap red.	ΔQALY	Gap red.
Utilitarian	0.2%	23.4%	1.4%	10.6%	1.7%	20.8%
Egalitarian	2.8%	28.4%	3.7%	19.5%	6.1%	28.0%
Sufficientarian	2.2%	18.3%	2.1%	8.8%	4.9%	23.0%
Prioritarian	0.3%	19.7%	1.7%	9.0%	2.3%	25.9%
Needs-Based	0.8%	13.3%	1.6%	11.3%	4.0%	20.9%

Table 2: Percent change in aggregate cumulative QALYs and in the Q4–Q0 QALY gap, relative to no intervention, under permanent targeted, systemic, and combined interventions, across allocation policies.

tions, this may provide an incomplete picture. Our results show that who is targeted is only one part of an intervention's effects. Timing and duration can also substantially change its long-term consequences, even when the targeting strategy is unchanged. An intervention introduced early may produce greater benefits than the same intervention introduced later, while a short intervention may have limited long-term effects even when it initially improves access.

This distinction is directly relevant to AI-mediated healthcare delivery. An algorithm used to identify patients for care management, prioritize outreach, or direct limited navigation resources does not itself remove the structural barriers that determine whether a patient ultimately receives care; its effect depends on what happens after the algorithmic decision is made. A patient correctly identified as high priority may still encounter transportation barriers, limited appointment availability, or other constraints that prevent the intended benefit from being realized. Evaluating an algorithm's predictive performance or cross-sectional fairness may fail to account for the healthcare infrastructure that determines how its recommendations are put into practice. A more complete evaluation should consider the full care pathway, from diagnosis, to delivering interventions, ensuring access to care, and assessing long-term health outcomes. Longitudinal simulation provides a framework for examining these processes over time and identifying points where disparities may persist, accumulate, or emerge. This approach can reveal potential sources of inequity and their implications for patient outcomes before an AI system is deployed in clinical practice.

Limitations and Future Directions

Our simulation is intended as a demonstration rather than a calibrated estimate of any real intervention's effect. The simplified representations of access, health progression, and intervention effects mean that the absolute QALY differences and timing thresholds should not be interpreted as empirical predictions. The model allocates from a fixed monthly budget, which likely drives the diffuse, policy-dependent shifts we observed in quintiles Q1, Q2, and Q3 (Figure 2). A different budget assumption (e.g., an expandable or elastic supply) could change or eliminate this effect, so it should be treated as a property of this model's resource assumption rather than a claim about targeted interventions. In addition, we evaluate one intervention strength for each intervention type, leaving broader sensitivity and robustness analyses for future work. Finally, while our simulation examines fairness through the lens of distributive justice, it does not address values such as autonomy, agency, transparency, accountability, and contestability, which concern the procedural and relational conditions under which healthcare decisions are made, experienced, and challenged. For simplicity, we represent access as a static, individual-level parameter, abstracting away institutional processes that govern resource allocation, patient categorization and prioritization, and how healthcare providers act on these categories (Whitehead 1992; Levesque, Harris, and Russell 2013). These processes may themselves be influenced by AI-mediated systems, which our model treats as fixed.

Future work should test whether the timing and duration effects observed here persist under empirically calibrated parameters and in other settings. Varying the budget assumption could distinguish effects driven by fixed resource constraints from more general properties of access-improving interventions. In addition, combining simulation with qualitative or participatory methods could help examine questions that the model cannot capture, including how patients and clinicians view trade-offs in the timing, duration, and targeting of access interventions. More broadly, this could help connect the structural dependencies made visible through simulation with values such as agency, participation, and contestability that are central to the evaluation of healthcare AI.

Conclusions

This work uses longitudinal microsimulation to examine how the effects of access-improving interventions depend on who they reach, when they begin, and how long they last. Our results show that these factors interact in ways not apparent from a single allocation decision. Modelling patient trajectories over a 50-year horizon revealed temporal dependencies and showed that a targeted intervention can have limited long-term effects when introduced too late or withdrawn too soon. More broadly, these results demonstrate how simulation can explore the consequences of intervention strategies before deployment. It can help identify when interventions have limited or uneven effects and distinguish the effects of access from other sources of inequality, such as underlying health and treatment effectiveness. Simulation is not a substitute for empirical evaluation or real-world monitoring, but provides a controlled setting for identifying potential failure modes and generating questions for further study. Evaluating healthcare AI therefore requires looking beyond who an algorithm allocates resources to and considering the conditions under which those resources become realized benefits over time.

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Preserving Function, Quality and Safety in Medical Process Automation

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Abstract

The automation of processes generally arises from a desire to improve their efficiency, quality or cost-effectiveness. However, due to technical limitations, these automated processes may not effectively perform the function of the original manual process. If a failure of functionality in an automated process is unknown to stakeholders this creates a risk of harm from unexpected behaviors. Therefore it is necessary to determine if an automation can achieve the desired functionality before implementing it in the place of an exiting manual process. This paper introduces a critical distinction between replacement of a manual process with an automated one, and automation of a manual process. In the latter the automated process maintains the function. In the former, a new automated process is used in place of the old manual process, but it does not perform the same function. We then develop a task-based analytic framework to assess process automation feasibility. We apply this framework to a case study of patient case summarisation software, to determine the extent to which the existing medical process can be automated using a large language model. Our analysis highlights three factors impacting automation feasibility: information availability, behaviour of the selected technology, and accountability requirements.

Introduction

Algorithmic process automation often brings with it the promise of efficiency, cost savings and objectivity. Recent developments in general purpose large-language model (LLM) technology have created opportunities across sectors for increased automation in processes requiring natural language processing, such as document creation, communication, and administration (Noy and Zhang 2023; Dell'Acqua et al. 2026; World Economic Forum 2023). The healthcare sector has sought to capitalise on such opportunities, with the implementation of tools like digital scribes (van Buchem et al. 2021), patient record summarisation (Oliveira et al. 2025), and medical chatbots (Hindelang, Sitaru, and Zink 2024).

However, alongside benefits, process automation also brings risk of harms (Shelby et al. 2023). These harms can occur due to inherent technical limitations of AI tools used for process automation (Raji et al. 2022). Failures of tool validation and communication of assumptions between stakeholders may then lead to the development of automated processes that give the appearance of functionality while obscuring these technical limitations, and the resultant risk of harms (Kernahan et al. 2025).

The increasing prevalence of general purpose AI tools heightens these risks even further. With general purpose AI tools a process automation does not need to be specifically designed, instead the AI is able to figure out the actions to take as it moves from one task to another along a process (Ye et al. 2023; Fan et al. 2025). This automation of automation creates greater opacity around automated processes, leading to greater risk that functional failures at the task level will be hidden from users and impacted stakeholders. Therefore, determining feasibility of automation is a critical prerequisite to implementing a process automation.

In this paper we make two contributions to help resolve this challenge. First, we define a distinction between process automation and process replacement, and introduce the idea of imposter automations. Second, we introduce a task based analytic approach to examine feasibility of process automation.

The paper is structured as follows. In the *Background and Theory* section we present our conceptualisation of process automation and feasibility. Then, in the *Case study* section we examine an example of a specific clinical process and assess its automation feasibility using a task-based analysis. Finally, in the *Conclusion* we summarise our contributions and highlight the need for future work in this space.

Background and Theory

The extent of automation of a Human-AI system is often conceptualised in levels, corresponding to the amount of control a human agent has compared to an AI agent (Vagia, Transeth, and Fjerdingen 2016; Ficuciello et al. 2019). At the lowest level the human agent has complete control and at the highest level the AI agent does. The levels in between form a gradient in which control is gradually passed from the human agents to AI agents. To decide the appropriate level of automation for a system, the capability of the human and AI agents need to be considered to determine the type of control that needs to be applied.

However, more complex autonomous systems involve

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multiple different tasks. A robotic surgeon, for example, involves a series of perception tasks, mechanical tasks, and analytics tasks strung together to form a long process chain. An autonomous robot may have varying capacity to perform each of these tasks and hence, a different level of control would be required for each, giving multiple different levels of automation within one system. Some tasks may be able to be fully automated, others may be necessary to keep completely manual, and some should be a mix.

Applying a single level of automation to every task, while ignoring these different levels of capacity and control requirements can hide failures of functionality and safety. Thus, the extent of automation of a process should not be considered as an overall level automation based on the type of control applied by humans. Instead it should consider how many tasks along the process chain can be automated, and which need to remain with human actors.

Process automation or process replacement

When considering the safety of AI in healthcare, there is often a focus on how harms can be mitigated by applying controls to the automated processes. Resulting in recommendations to perform bias mitigation (Obermeyer et al. 2019; Buolamwini and Gebru 2018) and expand monitoring and oversight (Goddard, Roudsari, and Wyatt 2012; Labkoff et al. 2024). While these safety mechanisms are important, there is a lack of consideration given to the fact that many harms arise because of the inherent inability for certain processes to be automated with AI.

Raji et al termed this the 'fallacy of AI functionality' (Raji et al. 2022). That is, there is an embedded assumption underpinning AI ethics and safety discourse, that AI technology works. However, in their paper Raji et al described multiple failure modes that could prevent AI tools from operating with the required functionality. One of these failure modes was 'impossible tasks', where one of the tasks intended to be conducted by the AI tool is conceptually or practically impossible to achieve. When a process is automated, these impossible tasks become embedded within a series of other automated tasks, while the overall process seems to behave as expected. This gives the illusion of functionality while obscuring the point of functional failure.

Therefore, we introduce a critical distinction between the automation of a manual process (process automation), and the replacement of a manual process with a different automated one (process replacement), see figure 1. In the former, the automated process maintains the function of the original manual process. In the latter, a new automated process replaces the old manual process, but it does not perform the same function. In some cases, it may be clear to users or impacted stakeholders that the replacement process does not perform the necessary function. However, in other cases it may be unclear that the replacement process is not performing the desired function. We term an automation which contains an unknown functionality failure an *imposter automation*.

Thus, before automating a more multi-task process it is critical to consider each task node within the process chain to determine which are possible to automate.

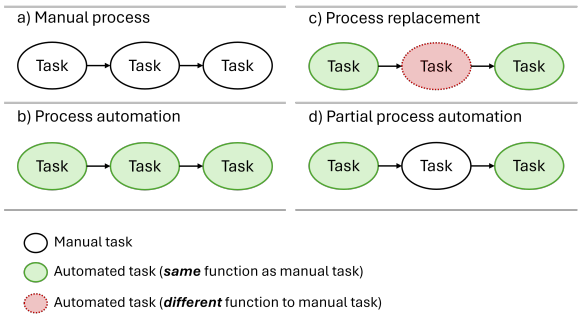


Figure 1: Different automation manifestations

Task-based automation feasibility analysis

We apply our conceptualisation of process automation feasibility to an example of process automation at a Dutch academic hospital. Our case study is the automation of a discharge letter summary writing process, using an LLM-based tool. We conducted interviews with the product development team and product users (7 interviews) of the discharge letter tool, reviewed relevant product documentation, and conducted a site visit to observe the tool.

A task-based analysis of automation feasibility was conducted in four phases. First, we mapped the key tasks in the manual process. Second, we outlined the functional requirements of the process. Third, we examined the how the limitations and capabilities of both the technology and the human actors, affected the ability to achieve these requirements. Finally, based on these limitations and capabilities we assessed each task as either *automation feasible* or *automation not feasible*.

Case study

Patients who are admitted to hospital for inpatient care receive a discharge letter at the end of their stay. A discharge letter summarises the key medical events that occurred, the medical care received and outlines information necessary for ongoing treatment or recovery. These letters may be provided to another hospital, medical specialists, and/or to the patient's general practitioner.

A patient discharge letter document is generated based on a template. The template includes a blank section where physicians manually input a summary of the course of the patient stay, based on the detailed clinician notes recorded throughout the admission. The process of manually creating a discharge letter summary consists of three main tasks (figure 2). Firstly, the summary is updated daily throughout the patient's admission to the hospital. Second, when it is time to discharge the patient, the summary for the discharge letter is reviewed and final updates are made by a physician or physician's assistant on the ward. This includes ensuring that the summary contains the most up-to-date patient information, and that this information is accurate. Finally, the discharge letter summary is reviewed and approved by a senior physician.

The reality of drafting and maintaining the course summary section of the discharge letter throughout the patient

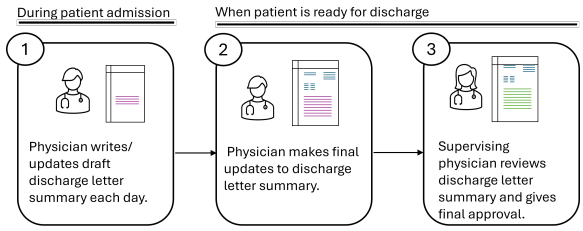


Figure 2: Three main steps in the manual discharge letter summary drafting process.

stay is a pain point in clinical practice. It is considered an administrative burden and is a low priority for physicians. As one explained:

> "At the end of the day, after handover at around 5:30pm, when you're ready to go home, that's the moment that you can add something to the letter. So, what we noticed - and what we have been noticing for the last 30 years or something - is that the making of the discharge letter was always a big problem."

This inconsistent maintenance of the discharge letter throughout the patient stay creates a high mental load on the physicians tasked with drafting the final discharge letter. The first step of the process, writing and maintaining the discharge letter summary, is not treated as a priority throughout admission. Therefore, there is often no letter drafted when it is time for the patient to be discharged. Physicians described it as a "mental disaster, that you've opened the charts, the [patient] is leaving, and there's nothing there, and you have to do everything from scratch at that moment."

The goal of automating the discharge letter summary creation process was therefore, to reduce this administrative burden on physicians, both by reducing the time they need to spend on the task, but also by reducing the mental load of carrying out the task.

Capabilities required to meet functional requirements

When discussing the requirements of the discharge letter summary tool, the clinicians interviewed made explicit that automation should not come at the expense of quality of the summary. That is the information in the final output should be as complete, relevant and accurate as if it had been drafted by the users themselves. One physician stated that the ideal summary should be "not too long, so that no one will read it, but it's not too short that it's limited in the amount of information". Additionally, there needed to be accountability for any errors in the summary.

We examine the capabilities of automation to preserve each of these functional requirements.

Requirement 1: The summary should be complete The data scientists worked with clinicians to understand what textual data they use to write the discharge letter summary. They ensured that those were provided as input into the discharge letter summary tool, noting that "the main thing that we used was whether they look at it when they write their own discharge letter." This is can be a large amount of free text data per patient "the system is fed with [. . .] the daily activities or the daily reports, everything that's written down in ultrasound investigations, but also on the routine care or the medications that is being given." All of this textual data is created by clinicians writing notes throughout the patient stay.

The information in these clinical notes is not always complete and this then affects the quality and completeness of the LLM generated discharge letter summary. As one clinican put it:

> "If the doctors aren't really putting the right information into their daily reports, then the AI will not be able to generate relevant discharge letters."

The clinicians indicated skepticism that the clinical notes being used as input into the summary tool would ever be of high enough quality that the summary output would be guaranteed to contain all the necessary information. Thus, there would always have to be a clinician reviewing and editing the output, impacting the potential automation of the second task.

Requirement 2: The summary text should be both relevant and error free Physicians reported that in general the LLM output in the discharge letter tool was better than expected, considering it a useful starting point for drafting the discharge letter summary.

No major errors in the LLM output were observed by the stakeholders we interviewed, however they did note that the text can sometimes be unnatural or misleading in the way events are phrased. One clinician noted that the style of writing in the LLM generated summaries could often be more literary than medical "sometimes it has like a funny way of writing things down, so it's more like literature, than it's medical language. That happens frequently."

All the clinicians interviewed mentioned that the LLM output will often contain minor hallucinations that make it seem as though there is causality between two unrelated events. One noted "it doesn't seem to really make up things... what does happen is that it sometimes links two things together that happened at the same time but aren't causally related". This is something that they are aware they must check and correct for in each letter.

There is also often superfluous information in the output. This occurs because of the way that the LLM prompt is structured to check for important low frequency events, which would otherwise be missed in the LLM summary. One of the senior clinicians involved in the development process noted:

> "Things that occur not that often is difficult, or seems to be be difficult, for the AI to really find [. . .] if there is only one patient with [a condition] and you're not specifically asking for it to look for that information, then it might skip that."

They explained that to mitigate this focus on common events, the system prompt explicitly prompts the LLM summary to report on potential low frequency, high consequence events. This prompt structure is necessary to minimise the

risk that important information is not present in the LLM generated summary, however, for patients where these rare events have not occurred the summary will state 'X did not occur' for each event "[the output] says the [patient] 'didn't have feeding disabilities', they 'didn't go on the ventilator', etc. I think that our prompt is a bit to blame for that."

These limitations in the quality of the output, combined with the limitations of the input data, mean that the LLM generated summary cannot be issued automatically for review to the supervisor but must first be edited by clinicians. Meaning that automation of the second task is not feasible. However, both data scientists and clinicians noted that this inconvenience is preferable to the risk of missing important information, with one clinician stating that "it's always better to have the information available that that you can delete, than that something is missing that was actually really relevant."

Requirement 3: There must be accountability for the process output In the previous manual process of creating the discharge letter, a physician would draft the patient summary and then the supervising physician would review that letter, make any necessary changes and then sign off on the letter. This review process firstly reduces the chance of errors in the information by enacting the "four eye principle" on every discharge letter. Secondly, it ensures there is accountability that can be attributed to the signing medical professional for what is written in the letter. Physicians who were engaged during the tool design process specified that the final task in the process, the review by the supervisor, needed to remain manual. This was to maintain the existing levels of accountability and safety control in the final letter check. Avoiding accountability gaps that may arise in automation was so important that hospital departments with fewer review steps in their discharge letter drafting processes did not implement the automation at all, for fear that their final check step would be at risk of automation. According to one of the data scientists:

> "We've also talked to departments where there is not a standard process with a double check by a supervisor before you use it. So those were generally more skeptical, which makes sense because we also don't want to just, generate something and then have that automatically being used in a letter"

Therefore, when designing the new process to include automation. The third task, supervisor review, should remain manual.

Feasibility of process automation

Our analysis highlighted three main conditions that impacted the feasibility of automating tasks in the discharge letter summary writing process: availability of necessary input data, behaviour of the selected AI technology, requirements of human accountability.

The first and second of these conditions relate to the capabilities of the technology. LLMs are able to summarise large amounts of information into a clearly structured piece of text. This meant that the tool was able to generate a first draft summary which provided a sufficient starting point for physicians to use. Hence, automation is feasible for the first task. However, information gaps in the input data, combined with the behavioural tendency to hallucinate mean that the tool did not have the capability to output a final summary that was both complete and accurate. Thus, it is not feasible to automate the second task.

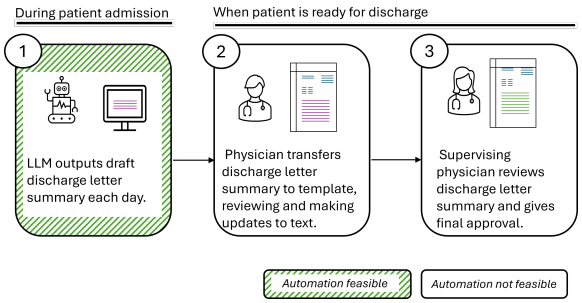


Figure 3: Only the first task within the discharge letter summary drafting process was feasible to automate.

The third condition identified in our analysis relates to the capabilities of the human actor in the process. In this case, it was not the capability to perform a task that mattered, but the capability to be accountable for having performed a task. Existing professional and legal obligations mean that task three must be performed by a human physician. Thus, it is not feasible to automate the third task.

Automation of either the second or third task would therefore be an instance of process replacement as the automation would not be performing the same functions of the original manual tasks. Therefore, the feasible extent of automation is a partial process automation, where the first task is automated and the second and third tasks remain manual (figure 3).

Conclusion

In this paper we have discussed the issue of functional failures in automated processes. To help articulate this issue, we drew a distinction between process automation and process replacement, and assigned the term *imposter automation* to occurrences of process replacement where the functional failures of the automated tasks are unknown. In doing so we have introduced language to name and discuss such phenomena.

Finally, we have shown how task-based analysis of automation feasibility can be used in the design process to determine the extent to which a manual process can be automated. This can ensure that the design does not result in unintended process replacement, and that the risk of implementing of imposter automations is avoided.

While further work is required to build out a full conceptualisation an analysis framework, this research provides a starting point for discussions about process automation feasibility analysis in healthcare.

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Context-Aware Hierarchical Bayesian Modeling of IVF Laboratory Environmental Conditions

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Abstract

IVF pregnancy rates are routinely modeled using patient-level variables, while high-resolution laboratory environmental data remain underutilized. We show that this is a missed opportunity. Rather than relying on raw sensor averages, we engineer 55 context-aware temporal features including rolling thermal stability, simultaneous temperature-humidity adherence, peak stress duration, and post-stress recovery speed that capture the dynamics of incubator microenvironments. On 61 weeks of data from an Asian IVF clinic, these features reduce cross-validated prediction error to 1.27%, compared to 3–5% for raw averages. We then train a hierarchical Bayesian Beta regression model that shares environmental effects across an Asian and a Northern European clinic via partial pooling, while preserving site-specific baselines. On held-out data from the Northern European clinic, the model achieves $R^2 = 0.86$ and a 64% error reduction for the 35–39 age group over a naive baseline, demonstrating that structured environmental monitoring contains clinically meaningful, transferable signal.

Introduction

The success of IVF depends on several interacting factors, including the patient's age, ovarian reserve, sperm quality, lifestyle, the stimulation protocol, embryo quality, etc (Younglai et al. 2005). In addition, the physical environmental conditions of the laboratory, like temperature, humidity, and air quality, play an important role. During the culture period, embryos are kept inside incubators that tightly control their internal environment (Mantikou et al. 2013; Varzaneh et al. 2025). However, these incubators do not operate in complete isolation. Each time an incubator door is opened, room air enters, and the time needed for the incubator to return to its optimal condition depends directly on how stable the surrounding laboratory environment is.

Despite this, environmental monitoring in IVF laboratories is often treated primarily as a compliance exercise: sensors trigger alerts when fixed thresholds are breached, yet the rich, high-resolution sensor data are rarely analyzed statistically. This represents a missed opportunity. Prior work has focused on patient-level predictors (Kupka et al. 2016), suggesting the influence of environmental and procedural factors. Complementary controlled experiments, such as (Swain 2012), demonstrate that even minor changes in culture conditions, such as oxygen levels, temperature stability, and media pH, can directly affect embryo development and clinical outcomes. However, these studies typically examine single variables under idealized conditions and do not capture the complex, time-varying fluctuations present in routine clinical practice. As a result, ignoring high-resolution environmental data may miss the cumulative effects of real-world stressors that help explain differences between laboratories.

We address this gap with two main contributions:

- **First**, we introduce *context-aware features* computed from sensor data that encode temporal patterns invisible to raw statistics: rolling thermal stability, the fraction of time temperature and humidity are simultaneously within their biological ideal ranges, the longest uninterrupted stress episode, post-stress recovery speed, and first week transfer-window summaries.
- **Second**, we develop a *hierarchical Bayesian Beta regression* model to analyze IVF outcomes across two clinics (Asian and Northern European). The model uses partial pooling to share information about environmental effects between the two sites, while allowing each clinic to maintain its own baseline level. In this way, data from the Asian clinic helps the model learn general and stable relationships between environmental conditions and IVF outcomes, which in turn improves prediction performance on the Northern European dataset without assuming that the two clinics are identical.

This study also addresses three key questions:

(i) Do context-aware environmental features outperform simple aggregated sensor measures?

(ii) Which features are most important for prediction, and are these effects consistent across the two clinics?

(iii) Does sharing information between clinics through partial pooling improve performance compared to a single-site baseline?

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Table 1: Environmental conditions and pregnancy rates: mean±std, training period.

Variable	Asia	Northern Europe	Unit
Environmental			
Temperature	23.4 ± 0.6	24.0 ± 0.6	°C
Humidity	37.2 ± 9.8	42.8 ± 8.3	%
CO_2	168 ± 22	189 ± 18	ppm
TVOC	148 ± 420	312 ± 380	ppb
Pregnancy rates by age group			
PR (<35 yrs)	44.1 ± 3.1	38.6 ± 18.4	%
PR (35–39 yrs)	26.6 ± 2.8	34.7 ± 12.1	%
PR (40+ yrs)	13.9 ± 1.8	16.2 ± 13.7	%

Data

Environmental sensor data were collected at two IVF laboratories at 10-minute intervals during 2024–2025. We refer to the Asian clinic as the **source** clinic because it provides substantially more labelled data (61 weeks) and serves as the basis for feature selection and partial-pooling priors; the Northern European clinic is the **target**, representing the site we aim to generalise to. Both stations record temperature (°C), relative humidity (%), CO_2 (ppm), and TVOC (ppb). Pregnancy rates were provided for patients under 35, aged 35–39, and aged 40 and above. Table 1 summarises the training period.

The Asian record spans January 2024–October 2025, interrupted by a 196-day collection gap caused by sensor data unavailability at the clinic; this splits it into two continuous segments of 19 and 42 weeks (61 total). The Northern European clinic provides 14 months (November 2024–December 2025). The large standard deviations in the Northern European data mostly come from months with only one to three patients in an age group, where the reported rates are driven more by small numbers than by environmental conditions.

Data Structure. Environmental variables are recorded every 10 minutes. Pregnancy rates were reported at different temporal granularities by the two clinics: weekly for Asia and monthly for Northern Europe, reflecting the format in which outcome data were made available to us by each laboratory. Environmental features are therefore aggregated to match each clinic's reporting interval (weekly for Asia, monthly for Northern Europe), so that features and outcomes are always aligned at the same resolution. The 196-day gap in the Asian data is handled by treating the two segments as independent time series during cross-validation, with fold boundaries never placed across the gap to avoid information leakage; the autoregressive term $\gamma \operatorname{logit}(y_{t-1,c})$ is reset at the start of each segment.

A further challenge is that outcomes at time T reflect conditions from roughly two to six weeks earlier. Since exact embryo-level timestamps are unavailable, we approximate this delay by including lagged environmental summaries from the previous period $(T-1)$, providing a practical alignment between past conditions and observed outcomes.

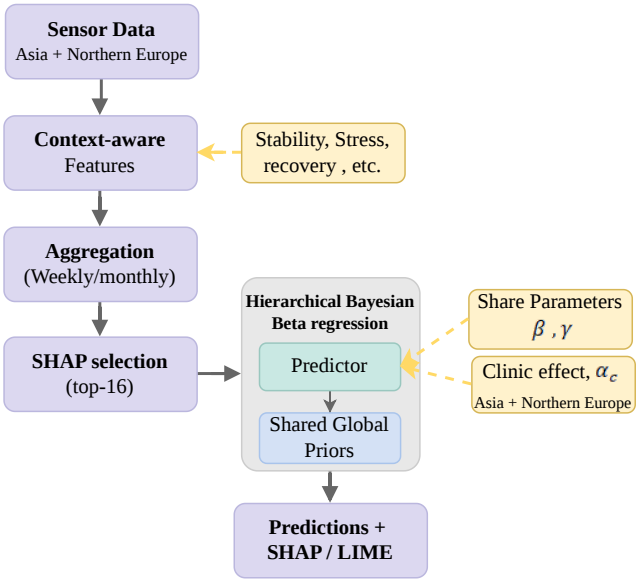


Figure 1: From sensor data to predictions. Context-aware features are extracted and aggregated over time, then fed into a hierarchical Bayesian model that shares information across clinics while preserving site-specific differences.

Method

The pipeline has four steps (see Figure 1): (1) context-aware feature engineering from 10-minute readings; (2) aggregation to weekly resolution for Asia and monthly for Northern Europe; (3) SHAP-based selection of the top-16 features per age group on Asian data to keep sampling tractable; (4) joint modelling of both clinics using a hierarchical Bayesian Beta regression.

Step 1: Context-Aware Features. For temperature T_t with optimum $T^* = 22.5\,°\text{C}$, the stress index is $s_t^T = \max(0, |T_t - T^*| - 1.5)/3$; an analogous expression governs humidity ($H^* = 42.5\%$, deadband $\pm 2.5\%$). The composite comfort index $c_t = (1 - 0.5\, s_t^T - 0.5\, s_t^H)^+$ summarises environmental quality. Beyond these pointwise quantities, we compute: the rolling 1-hour temperature standard deviation σ_t^T (and its period mean and stable fraction below 0.5 °C) as a measure of thermal stability; the ideal-zone adherence $z_t = \mathbf{1}[T_t \in [21, 24]°\text{C}] \cdot \mathbf{1}[H_t \in [40, 45]\%]$ capturing simultaneous optimality; the longest consecutive stress episode; the recovery score $r = \mathbb{E}[c_{t+1} \mid c_t < 0.5]$. Short-term lags (1–3 hours) capture local dynamics, while summaries from the preceding period $(T-1)$ provide coarse temporal alignment with outcomes. The final feature set includes 55 context-aware variables per period.

Step 2: Temporal Aggregation. Asia segments are aggregated to weekly resolution independently; the Northern European data to a monthly resolution. This yields 55 features per period.

Step 3: SHAP Feature Selection. Given the small number of Northern European training months, using all features would lead to an unstable Bayesian model. We therefore

train an XGBoost model (Chen and Guestrin 2016) on the Asian data and select the top-16 features based on mean absolute SHAP importance for each age group (Lundberg and Lee 2017). This reduces dimensionality while retaining the most informative predictors.

Step 4: Hierarchical Bayesian Beta Regression (Bhuwalka et al. 2023). Let $y_{t,c} \in (0,1)$ denote the pregnancy rate for clinic c at time t. Since the outcome is a proportion bounded between 0 and 1, we model it using a Beta distribution:

$$y_{t,c} \sim \mathrm{Beta}(\mu_{t,c}\phi,\ (1-\mu_{t,c})\phi). \tag{1}$$

The expected value $\mu_{t,c}$ is linked to the predictors through:

$$\eta_{t,c} = \alpha_c + \mathbf{x}_t^\top \boldsymbol{\beta} + \gamma\,\mathrm{logit}(y_{t-1,c}), \quad \mu_{t,c} = \sigma(\eta_{t,c}). \tag{2}$$

As shown in Eq. (2), α_c is a clinic-specific intercept that captures baseline differences in pregnancy rates between clinics. The coefficient vector $\boldsymbol{\beta}$ is shared across clinics and represents the common effect of environmental features. The autoregressive term $\gamma\,\mathrm{logit}(y_{t-1,c})$ accounts for temporal dependence, reflecting that current outcomes are influenced by previous ones.

To enable partial pooling across clinics, we place a hierarchical prior on the intercepts:

$$\alpha_c \sim \mathcal{N}(\mu_G, \sigma_G). \tag{3}$$

where μ_G is informed by approximate age-specific clinical pregnancy rates from the general IVF literature (Appendix A), and σ_G controls how much clinics may deviate from this shared baseline. σ_G is inferred from data, yielding automatic partial pooling without a fixed pooling decision.

We use weakly informative priors for the remaining parameters in Eq. (4). Posterior inference uses NUTS (Hoffman and Gelman 2014) in PyMC (Salvatier et al. 2016) (2 chains, 1500 warm-up + 1500 draws, target acceptance 0.95).

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, 0.09I), \quad \gamma \sim \mathcal{N}(0.3, 0.09), \quad \phi \sim \mathrm{Gamma}(3, 0.1) \tag{4}$$

Results

Setup. Northern European months 1–11 are used for training; months 12–14 (October–December 2025) are held out. The decision to use Asia as source and Northern Europe as target rather than a location-independent split is motivated by the asymmetry in available labelled data: the Asian clinic provides 61 weeks sufficient for cross-validation, while the Northern European clinic has only 14 months in total. A location-independent split would mix sites within folds, preventing evaluation of cross-site transfer, which is the central claim of this work. We compare the hierarchical model (HM) with a single-site XGBoost baseline (XGB) and a naive predictor (train-mean). All results use the held-out test set only. The models are evaluated using CV-MAE, MAE, and R^2.

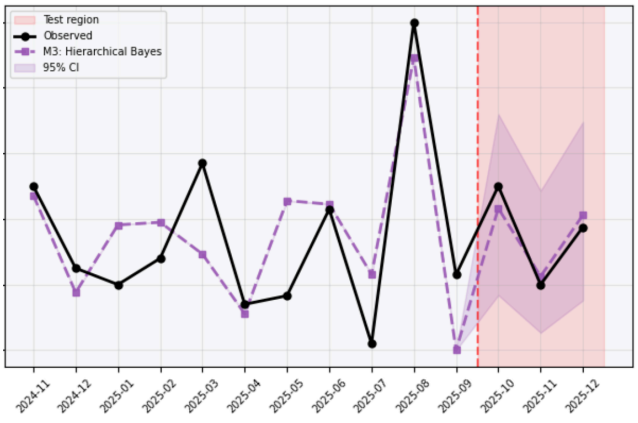


Figure 2: PR 35–39: observed (black) vs. HM posterior mean (dashed). Shading = 95% credible interval. Pink = test period.

Table 2: Asian clinic — 5-fold time-series CV (61 weeks). Context-aware vs. raw-feature baseline (in parentheses).

Age group	Model	CV-MAE	(Raw)
PR <35 yrs	RF	1.25%	(1.94%)
PR 35–39 yrs	RF	0.85%	(1.57%)
PR 40+ yrs	RF	1.30%	(1.83%)

Asian Cross-Validation. Table 2 reports 5-fold time-series cross-validation on the Asian data. Context-aware features achieve CV-MAE of 0.85–1.30%, compared with 1.57–1.94% when using only the four raw monthly statistics (mean temperature, humidity, CO_2, TVOC). This consistent improvement across all age groups indicates that temporal patterns provide useful predictive information beyond simple averages.

Northern European Test Results. Table 3 and Figure 2 report test-set metrics.

HM achieves MAE $= 4.30\%$ and $R^2 = 0.86$ for the 35–39 group, a 64% improvement over naive. This group is the most reliable target because it has the largest per-month patient count at both clinics, making the observed rate a stable estimator. We caution that with only 3 test months the reported point estimates carry substantial uncertainty; these results should be interpreted as a promising preliminary signal rather than a definitive claim.

The strongly negative R^2 values for XGBoost (-5.09, -0.28, -0.33) indicate that a point-estimate model without regularisation overfits the 11 training months and collapses on the test set as a direct consequence of the limited sample size. This pathological behaviour is precisely what motivates the hierarchical Bayesian approach: the informative prior and partial pooling act as regularisation, preventing overfitting in the low-data regime. For the 40+ group HM recovers $R^2 = +0.08$ and a 17% improvement; the hierarchical prior stabilises predictions against counting noise where the single-site baseline ($+1\%$) cannot. The under-35 group is noise-dominated: monthly rates reach 0% or 100%

Table 3: Northern European clinic — test-set results (3 held-out months). ΔN = improvement over naive. **Bold** = best MAE per group.

Age group	Model	MAE	R^2	ΔN
PR <35 (noise-limited)	Naive	8.76%	—	—
	XGB	17.05%	−5.09	−95%
	HM	**16.55%**	−8.81	−89%
PR 35–39 **(primary)**	Naive	11.80%	—	—
	XGB	11.80%	−0.28	0%
	HM	**4.30%**	**+0.86**	**+64%**
PR 40+ (few patients)	Naive	20.14%	—	—
	XGB	19.85%	−0.33	+1%
	HM	**16.71%**	**+0.08**	**+17%**

when only one or two patients are treated, and both models fall below naive, confirming that no environmental signal can be recovered at this level of aggregation.

Feature importance results from SHAP and LIME are presented in Appendix B. Temperature and CO_2 emerge as the most consistently important variables across both clinics, while TVOC shows a stronger negative effect in Northern Europe.

Limitations. The dataset is limited in size, particularly for the Northern European clinic (3 test months), which restricts statistical power and means reported metrics carry substantial uncertainty. Outcomes are aggregated proportions without patient-level information; factors such as embryo quality, stimulation protocol, and patient history are unobserved and may confound the estimated environmental effects. Similarly, clinic-month-level outcomes may correlate with staffing patterns, lab protocol changes, or seasonal effects unrelated to sensor readings.

Conclusion and Future Work

Context-aware features capture patterns that are not visible in simple monthly averages, and this is supported by results on the Asian data and the transfer to Northern Europe. The hierarchical model with partial pooling reaches $R^2 = 0.86$ for the 35–39 age group, where CO_2 and temperature appear consistently important across both clinics. The 40+ group shows only small improvements, and the under-35 group is mostly dominated by noise at this level of aggregation. Overall, the results suggest that routine environmental monitoring is critical to contain useful information about laboratory conditions that goes beyond simple threshold alarms. Future work should focus on larger datasets and adding patient-level and incubator-level data to get more reliable and biologically meaningful results. In addition, counterfactual analysis could be applied in this context to explore "what-if" scenarios, enabling the assessment of how changes in environmental factors (e.g., CO_2 levels or temperature) might influence success rates.

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Appendix A: Prior Specification and Beta-Binomial Extension

Choice of prior means μ_0. The global mean μ_G in Eq. (3) is centred on an approximate prior pregnancy rate μ_0 for each age group. These values are not directly tabulated in any single source; they are weakly informative approximations consistent with the ranges reported across European ART registries (clinical pregnancy rates per fresh autologous cycle) (Kupka et al. 2016). Because σ_G is inferred from data and both Asia and Northern Europe contribute many observations, the posterior is not sensitive to moderate changes in μ_0. Table 4 summarizes these starting points.

Table 4: Weakly informative prior means μ_0 used for μ_G, consistent with European ART registry ranges (Kupka et al. 2016).

Age group	μ_0	σ_0	Rationale
$<$35 years	0.40	0.10	Approx. clinical PR per cycle, European avg
35–39 years	0.28	0.10	Lower success, wider uncertainty
$\geq$40 years	0.12	0.08	Marked decline; narrow prior, broad posterior

Beta-Binomial extension. When monthly patient counts $n_{t,c}$ are available, replace Eq. (1) with $k_{t,c} \sim \text{BetaBinomial}(\mu_{t,c}\kappa, (1-\mu_{t,c})\kappa, n_{t,c})$, where $k_{t,c}$ is the number of clinical pregnancies and $\kappa \sim \text{Gamma}(2, 0.1)$. This weights each month by how many patients contributed—the statistically correct treatment for small-count subgroups, and the most important modelling improvement available for future work.

Appendix B: SHAP and LIME Interpretation

Figures 3 and 4 provide two complementary views of how environmental conditions relate to pregnancy rates (Salih et al. 2025).

The SHAP analysis (Figure 3) shows which variables matter most and in which direction across both clinics. In the Asian data, higher CO_2 values are associated with positive SHAP contributions. In contrast, in the Northern European clinic, higher TVOC shows a clear negative effect, and both CO_2 and temperature tend to contribute negatively. The opposite directional effect of CO_2 between sites is consistent with the model structure: β is shared, but the posterior is weighted toward the Asian clinic, which provides far more observations. This site-dependency in CO_2 effects likely reflects genuine differences in incubator types, ventilation systems, or CO_2 calibration protocols between the two laboratories, and warrants investigation with larger datasets. Temperature shows a more consistent pattern across sites.

LIME (Figure 4) provides a local view by explaining individual predictions. Averaged over Northern European months, lower CO_2 levels consistently contribute positively, while higher temperature and TVOC reduce predicted pregnancy rates. Higher humidity shows a positive contribution.

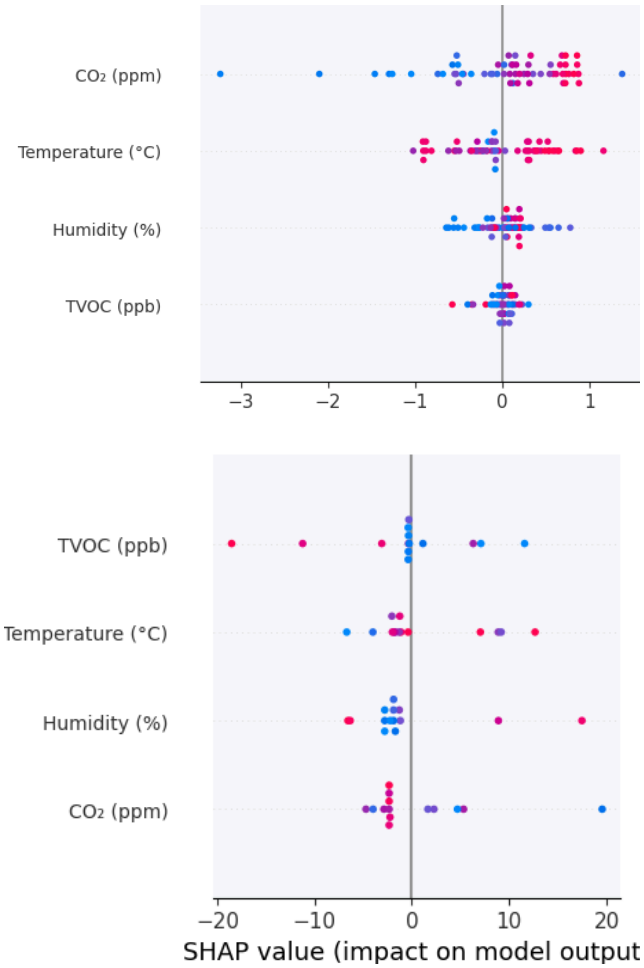


Figure 3: SHAP beeswarm for PR 35–39. *Top*: Asia (61 weeks). *Bottom*: Northern Europe (14 months). Red = high feature value; blue = low.

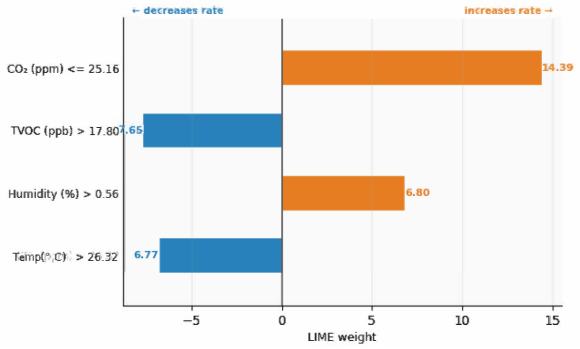


Figure 4: LIME explanation for December 2025, PR 35–39 (Northern European clinic).

Beyond Personalization: Governing the Care-to-Learning Boundary in Embodied AI

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Abstract

Embodied AI is entering long-term care not only as an assistive tool but as a participant in everyday social, recreational, and cognitive activities. Through these interactions, robots can learn from older adults' preferences, corrections, gestures, speech patterns, and behavioral responses. Existing work largely treats such learning as personalization, where information from a user improves future interaction with that same user. Governance changes, however, when interaction-derived learning is retained, aggregated, or reused to improve robots or models beyond the individual care relationship. This position paper introduces the care-to-learning boundary: the point at which situated care interaction begins to generate transferable value for broader AI development. Using robot-assisted tabletop games in institutional elder care as a running example, we distinguish four levels of interaction-derived learning and four practical indicators for locating the boundary. We argue that the transition across this boundary is graduated rather than binary and should trigger progressively stronger requirements for purpose separation, non-penalized refusal, and proportionate reciprocity.

1 Introduction

Imagine an older adult playing a tabletop game with a robot in a care facility. She reaches for a game piece; the robot misreads the gesture. She corrects it, repeats the movement, and explains what she intended. If the robot retains that correction and responds more accurately the next day, the interaction is naturally described as personalization. Now suppose instead that the same correction is stored, pooled with those of other residents, and eventually used to improve how future robots interpret older adults' gestures. What, exactly, has changed?

This paper focuses on everyday institutional care rather than diagnosis or clinical decision-making. Here, embodied AI means physically situated AI systems that sense, act, and adapt through ongoing interaction; personalization means adaptation intended primarily to improve future interaction with the same individual, rather than generalized improvement for other users or deployments. Even outside clinical contexts, such systems can shape wellbeing, agency, dependency, participation, and privacy. Robot-assisted board games have been evaluated with older adults with mild cognitive impairment in long-term-care facilities and day-care centers (Lin et al. 2025); assistive social robots can learn individual activity preferences from explicit and implicit feedback (Maroto-Gómez et al. 2025); and older adults have participated in robot design (Irfan, Kuoppamäki, and Skantze 2024) and robot-literacy initiatives (Ahtinen et al. 2025), while expressing concerns about privacy and control over learned data (Irfan, Kuoppamäki, and Skantze 2024).

Existing research thus recognizes several parts of this relationship. Human-robot interaction research examines personalization; participatory work treats older adults as active participants rather than passive recipients; and long-term-care robotics research identifies concerns such as consent, inequitable access, infantilization, and substitution of human care (Hung et al. 2025). Less visible is a further transition: when information generated within a care relationship stops serving primarily that relationship and becomes reusable input for broader AI development.

We refer to this transition as the care-to-learning boundary. It is not crossed merely because a robot processes information or adapts: interpreting a gesture, remembering a resident's preferred game, or adjusting speaking pace can remain within personalization. Governance begins to change when learning persists beyond the individual relationship, benefits others, travels across users or settings, or enters a pipeline for training, fine-tuning, evaluation, or generalized model improvement.

The boundary can be summarized as the shift from learning for the resident to learning from the resident for others. It is not a substitute for purpose limitation or a legal test of secondary use; it is a sociotechnical lens for identifying when adaptation embedded in a care relationship begins to create external learning value, including after interaction traces are aggregated or transformed. Crossing the boundary is neither necessary nor sufficient to establish unlawful processing; privacy, safety, and data-protection obligations may arise at every level.

This distinction is especially salient in institutional elder care, where personalization occurs within ongoing

relationships of support and institutions mediate access to technology and activities. Refusal may therefore carry practical social or service costs. Two errors follow: treating residents only as vulnerable data subjects can obscure their agency and capacity to teach, correct, and contribute, while treating ordinary care participation as automatically available for AI development converts support relationships into unrecognized sources of development value.

This paper makes three contributions: it distinguishes personalization from transferable model learning; makes the transition observable through four learning levels and four indicators; and proposes a graduated response based on purpose separation, non-penalized refusal, and proportionate reciprocity.

The aim is not to turn everyday care into a marketplace for data contributions or treat every interaction as compensable. The care-to-learning boundary identifies when ordinary personalization governance is no longer sufficient.

2 From Personalization to Transferable Learning

Learning from interaction is already familiar in assistive and social robotics. Explicit feedback includes stated preferences and corrections; implicit feedback is derived from interaction patterns. Maroto-Gómez et al. (2025), for example, show how an assistive social robot combines both forms of feedback to learn individual activity preferences. Such learning typically serves the same user who generated it.

Care interactions can generate learning that travels further. During a tabletop game, a robot may encounter ambiguous gestures, hesitation, corrections, or explicit demonstrations. Such signals may support the current activity but can also feed persistent adaptation and broader model improvement. We distinguish four levels as a functional governance taxonomy, not empirically observed stages of robot learning. Continual-learning research asks how robots learn from ongoing data streams (Lesort et al. 2020); our taxonomy instead tracks where interaction-derived adaptation is retained, who benefits, and whether it is reused beyond the originating relationship.

L0 — Transient interaction. Information is processed to complete the immediate activity but is not retained for later adaptation.

L1 — Personal adaptation. Information is retained primarily to improve future interaction with the same resident.

L2 — Shared or local learning. Learning derived from one resident begins to improve interactions with other users or systems within a shared care environment.

L3 — Transferable model learning. Interaction-derived information enters a wider development pipeline through aggregation, evaluation, benchmarking, training, fine-tuning, or other generalized model improvement. EDPB-commissioned technical training material notes that inputs and outputs generated during AI-system use may later become material for retraining or fine-tuning (Glerean 2025).

The care-to-learning boundary lies between learning that remains principally within the individual care relationship and learning that creates transferable value beyond it. The transition across it is graduated rather than binary: the transition may begin at L2 and becomes clearest at L3, while the case for stronger governance requirements strengthens as learning becomes more persistent, transferable, and externally valuable. The framework is heuristic rather than a strict taxonomy; benefit and development use may vary independently, and a deployment may sit at different levels for different data streams.

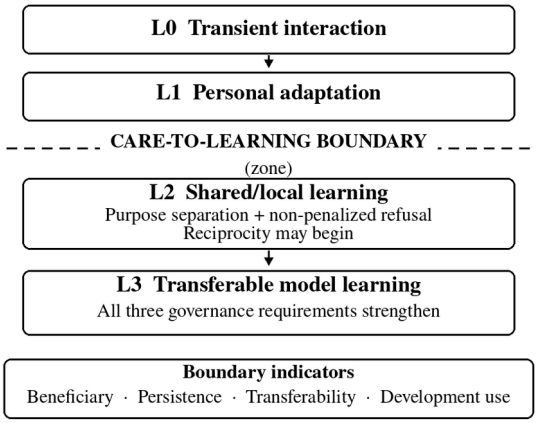


Figure 1. The care-to-learning boundary marks the transition from learning primarily for the resident to learning that creates transferable value beyond the individual care relationship. The transition may begin at L2 and becomes clearest at L3.

The boundary is functional rather than purely technical. Cloud storage does not by itself establish that it has been crossed, and local processing does not establish that it has not. What matters is what the learning is retained for, who benefits, and whether the resulting knowledge can travel beyond the relationship that generated it. Four practical indicators help locate the boundary:

Beneficiary. Who is expected to benefit? Adaptation that primarily improves the contributing resident's experience remains close to personalization; learning that benefits other residents, institutions, developers, or future customers has acquired external value.

Persistence. Does learning disappear with the session, remain within an individual profile, or persist as part of a reusable dataset, model update, or other resource?

Transferability. Can what has been learned be separated from the original person and reused across users, robots, institutions, or settings? Remembering that one resident likes chess differs from extracting a generalized pattern for interpreting certain movements.

Development use. Does the interaction enter training, fine-tuning, evaluation, benchmarking, or another process intended to improve systems beyond the ongoing care relationship?

These are design and audit questions, not validated metrics. Their purpose is to expose a transition residents may never see: the same tabletop game can look identical whether a correction disappears after the session, personalizes one robot, improves several robots in the facility, or contributes to a commercially deployed model. The indicators need not move together: persistent information may remain within L1 when it continues to serve only the same resident, while locally retained interaction data may nevertheless support L3 if derived updates enter generalized model development.

At L2, purpose separation and non-penalized refusal become relevant because one resident's interaction begins to benefit others, although reciprocity may remain modest or community-based. At L3, where interaction-derived learning enters generalized development, the case for distinct authorization, stronger refusal protections, and visible reciprocal benefit becomes substantially stronger.

3 Governing the Care-to-Learning Boundary

Crossing the care-to-learning boundary need not make broader learning impermissible. Older adults may willingly contribute to improving robots, and care institutions may participate in research and development that benefits future users. The problem arises when the transition remains implicit. The levels above are analytical; the three principles below are normative proposals rather than empirical findings. Each responds to a change created by movement across the boundary: purpose separation to a change in use; non-penalized refusal to an additional contribution role that should not be silently bundled with care; and proportionate reciprocity to value flowing beyond the relationship that generated it. They are not exhaustive, but address authorization, voluntariness, and value distribution with increasing force as learning crosses the boundary.

3.1 Purpose Separation

Participation in care should not silently authorize participation in broader model development.

A resident who agrees to use a social robot may reasonably expect the system to process information needed for the activity and, depending on the service, remember preferences that improve future interaction. These expectations do not necessarily extend to reusing interaction traces to train, fine-tune, evaluate, or improve systems for unrelated users. This gap is not merely hypothetical: in co-design work on companion robots, older adults raised concerns about who can access interaction data and whether it could be used beyond their consent, and the authors' design recommendations further caution that cloud-based processing may expose data to third parties that train on user data (Irfan, Kuoppamäki, and Skantze 2024). Learning from the resident for others should therefore follow a distinct and transparent authorization process rather than being silently folded into ordinary care.

Existing data-protection governance distinguishes AI development and deployment and recognizes that they may involve separate purposes and constitute separate processing activities requiring case-by-case assessment (European Data Protection Board 2024), while EDPB-commissioned technical training material notes that data arising during system use may later support training or fine-tuning, raising corresponding questions of transparency, consent management, and legal basis (Glerean 2025). The care-to-learning boundary adds a relational dimension: an interaction initially embedded in a relationship of support is transformed into an input to AI development.

Where reasonably feasible, system architecture should allow personalization without making transferable learning inseparable from ordinary use. Residents should be able to understand when the robot merely remembers them and when broader systems learn from them.

3.2 Non-Penalized Refusal

Purpose separation is meaningful only if refusal is practically possible.

Residents of long-term-care institutions may depend on the institution for social activities, companionship, assistance, and access to technology. A formally voluntary contribution can therefore become difficult to decline when transferable learning is bundled with valued care or meaningful participation. Declining participation in transferable learning should not, by itself, reduce access to ordinary or essential care or meaningful social activities.

Non-penalized refusal does not require every optional research activity to remain available unchanged to non-participants. A project explicitly designed to gather demonstrations for robot development may legitimately make contribution part of participation. The relevant contrast is between an activity transparently presented as research or contribution and an ordinary care interaction whose development function is hidden or bundled into the service.

Age or residence alone should not be treated as evidence of diminished decision-making capacity; where decision-

making support is needed, authorization and refusal mechanisms should be adapted accordingly. Nor should older adults be barred from contributing simply because they reside in care settings. Empirical ethics research in long-term care emphasizes both consent and the importance of residents' own voices (Hung et al. 2025). Safeguards should protect meaningful participation, not eliminate it.

3.3 Proportionate Reciprocity

Authorization and privacy are not the end of the governance inquiry. Authorization determines whether a use may occur; it does not determine how value created through that use should be distributed.

When a robot remembers a resident's preferences mainly to improve that resident's experience, personalization itself supplies much of the benefit. As learning becomes transferable, corrections may improve systems for others, reduce development costs, or enter commercial models. WHO's health-AI guidance notes that benefits from health-data sharing may include access to resulting technologies and investment in digital infrastructure, research capacity, and training (World Health Organization 2021).

Related patient-partnership work likewise stresses recognition of patient partners' experiential contributions (Smith, Bélisle-Pipon, and Resnik 2019). We therefore propose proportionate reciprocity: it should scale with the external value and transferability of learning, while its form and recipient should reflect where the contribution was generated. Ordinary personalization may require little beyond the improved service itself; shared or transferable learning creates a progressively stronger case for reciprocal benefit.

Reciprocity need not flow only to individuals, but resident benefit should remain visible rather than accruing only to institutions and vendors. Individually attributable contributions may justify recognition, co-design opportunities, or access to non-essential improvements; collective contributions may be reciprocated through resident-community programs. Institutional support may justify service credits, training, or infrastructure support, but should not substitute automatically for resident or community benefit when residents generated the learning value.

Two limits apply. Reciprocity does not require pricing every gesture, correction, or minute of interaction or turning care into a marketplace for data. Nor should any form of reciprocity make essential care conditional on contribution: valuable services must not be withheld to make participation incentives more attractive.

Taken together, the principles provide a graduated response: purpose separation addresses authorization, non-penalized refusal protects voluntariness where dependency may weaken it, and proportionate reciprocity addresses value distribution once learning becomes transferable.

4 Implications and Conclusion

The care-to-learning boundary carries implications for care institutions, developers, and older adults alike.

For care institutions, approving a robot for resident activities should not authorize every downstream use of the interactions those activities produce. Because institutions mediate access, dependency, and trust, they should distinguish ordinary personalization from broader learning and scrutinize arrangements through which developers obtain reusable interaction data.

For developers, information required for immediate operation or individual personalization should, where feasible, remain distinguishable from data retained for shared or transferable learning. Developers should be able to explain when information leaves an individual profile, how it feeds into generalized improvement, and who benefits.

For older adults, the framework steers between two errors: treating residents solely as vulnerable subjects requiring protection, and treating their capacity to contribute as permission to convert ordinary care participation into unrecognized AI-development value. Crossing the boundary may place an older adult in an additional functional role as a contributor to AI development without diminishing existing status or protections as a care recipient and data subject.

This paper offers a position framework, not an empirically validated classification. Future work should examine how residents, care institutions, and developers understand the transition across technical architectures. Although our focus is institutional elder care, the boundary may arise wherever personalized interaction becomes reusable input for broader model development. Its governance implications remain context-sensitive: institutional dependency strengthens the case for non-penalized refusal in long-term care, while other settings may involve different power asymmetries and forms of reciprocal benefit.

As embodied AI grows more adaptive, governance must attend not only to what a system learns about a person, but to when it begins to learn from that person for others, and to where the resulting value travels.

AI-Assisted Editing Disclosure

Generative AI tools were used only for language editing, proofreading, and formatting assistance. All substantive claims, arguments, citations, and final text were reviewed and approved by the authors, who take full responsibility for the content.

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More than Benchmarks: Machine Learning Competitions as Sociotechnical Systems

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Abstract

Machine learning (ML) competitions have become an established mechanism for developing and evaluating ML models, yet questions of fairness have largely focused on the datasets and algorithms produced through these competitions. Less attention has been paid to how competition design shapes the conditions under which people participate.

We approach ML competitions as sociotechnical systems and focus on how evaluation design, specifically moving from single-metric to multi-metric evaluation, may shape participation dynamics. This paper develops this perspective as part of the CHEETAH project. By connecting questions of evaluation with questions of participation, we seek to broaden how fairness in health AI developmnent is understood beyond the models that competitions produce.

Disclaimer: this is a working paper, and represents research in progress. We welcome contributions from the community, for comments or questions please email us at betsae@itu.dk or vech@itu.dk

Background

Every year, hundreds of machine learning competitions are organised to develop and compare solutions to problems in healthcare. In medical image analysis, competitions organised in conjunction with international conferences such as the Medical Image Computing and Computer Assisted Intervention (MICCAI) and the IEEE International Symposium on Biomedical Imaging (ISBI) have become an established mechanism for comparing algorithms and measuring progress (Maier-Hein et al. 2018; Eisenmann et al. 2023b). Winners are celebrated, with leaderboards becoming reference points for subsequent research (Eisenmann et al. 2023a,b; Orr and Kang 2024). Indeed, competitions have become a standard mechanism in ML practice. As Orr and Kang (2024) observe, the pervasive assumption in the community seems to be: "*how else would machine learning systems be evaluated and technological progress measured*?"

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Historically, new methods were often evaluated using authors' own datasets, making fair comparisons between solutions difficult (Maier-Hein et al. 2018; Raji et al. 2021; Orr and Kang 2024). Without a common basis for comparison, claims of superiority were difficult to verify, contributing to what Price (1986) caricatured as an endless cycle of "*anything you can do, I can do better*". The introduction of competitions or challenges, therefore, offered a fundamental change in how ML methods could be compared. By providing common datasets and evaluation protocols, competitions created a shared basis against which resulting solutions could be compared (Maier-Hein et al. 2018; Bron et al. 2022; Orr and Kang 2024). In doing so, they helped establish common standards that enabled a distributed community of practice to form and persist across space and time (Orr and Kang 2024).

A common way of organising evaluation is through leaderboard rankings based on a single score (Wang, Hertzmann, and Russakovsky 2024). In many competitions, this score is determined by a single evaluation metric, such as the Dice Similarity Coefficient for image segmentation or accuracy for classification (Maier-Hein et al. 2018; Bron et al. 2022; Reinke et al. 2024). This single-metric approach is widely adopted because it collapses complex, multi-dimensional model performances into a straightforward, linear ranking. By reducing evaluation to a single numeric value, organisers can unambiguously declare a winner, while participants are provided with a clear, mathematically tractable target for optimisation (Orr and Kang 2024).

Concerns about evaluation design. However, concerns have increasingly been raised about whether these simplified evaluation practices can adequately assess whether AI solutions are clinically safe, robust, and fair (Maier-Hein et al. 2018; Raji et al. 2021; Reinke et al. 2025). Standard metrics such as accuracy can remain deceptively high even when an algorithm is not clinically useful, while models optimised for benchmark performance may exploit superficial features in the data that improve reported performance but fail to generalise beyond the benchmark (Reinke et al. 2024; Jiménez-Sánchez et al. 2023). Moreover, once a metric becomes the basis for competition, it can also become a target in its own right, creating incentives to optimise for, or potentially "game", the metric rather than advance the underlying

task (Thomas and Uminsky 2022; Burnell et al. 2023; Orr and Kang 2024).

Representativeness of datasets and participants. Concerns have also been raised about the representativeness of the datasets used in competitions. For example, recent analyses of medical imaging challenges reveal substantial geographical and thematic imbalances in the data used, raising questions about how far leaderboard performance translates across different clinical populations and healthcare settings (Raji et al. 2021; Reinke et al. 2025). Beyond the datasets representation, research suggests that the design of ML competitions may have implications for who is able to participate and compete effectively. Success can depend on access to computational resources (Eisenmann et al. 2023b; Besiroglu et al. 2024; Hooker 2020), while the infrastructures through which competitions are organised can shape the conditions under which participants develop and evaluate their solutions (Luitse, Blanke, and Poell 2025).

These challenges raise fundamental questions about how ML competitions can be designed and evaluated to support both clinically robust algorithms and a diverse and inclusive participation. We address these questions within the broader, multi-year CHEETAH (Challenges of Evaluating Teams and Algorithms) project, funded by the Novo Nordisk Foundation Data Science Ascending Investigator grant. The project is structured around three key research questions:

1. RQ1: How should we evaluate ML algorithms with regard to robustness? How do we measure that the evaluation data is diverse enough, and how do we take into account performances on different subgroups of patients?
2. RQ2: How similar are the algorithms developed in a competition, and how could we optimise their robustness collectively and reduce the overall cost?
3. RQ3: How do different competition designs affect who participates and persists, and how does this affect who enters or stays in the data science field?

The first two questions address the data and evaluation infrastructures through which algorithms are developed and assessed. The third turns attention to the people who participate in these competitions and the conditions under which they are able to do so. In this abstract, we focus on this latter dimension, which we develop in the sections that follow.

Fairness beyond datasets and evaluation metrics

While research has highlighted important limitations and biases in competition datasets and evaluation metrics, considerably less attention has been paid to the competition settings in which participants develop and submit their algorithms. Emerging research nevertheless points to the importance of these settings within the broader AI development pipeline. Luitse, Blanke, and Poell (2025), for example, show how the platforms, organisers, participants, and infrastructures surrounding medical imaging competitions shape the conditions under which models are produced and evaluated. Participants in turn, develop strategies to navigate not only the technical challenge posed by the competition but also the particular arrangements through which that clinical problem is defined, evaluated, and ranked.

The compute divide. Competitions can also place unequal demands on participants. Marginal gains on competition leaderboards can require substantial computational resources, including access to GPUs and cloud infrastructure. More broadly, the growing computational demands of AI research have produced a "compute divide", with access to these resources increasingly concentrated among better-resourced organisations (Hooker 2020; Ahmed and Wahed 2020; Besiroglu et al. 2024). In a survey of biomedical imaging competition participants, Eisenmann et al. (2023b) reported that limited infrastructure was a bottleneck for a substantial proportion of participants. Luitse, Blanke, and Poell (2025) similarly show how competition infrastructures can facilitate broad participation while concentrating important resources and decision-making power among a small number of actors. Competitions may therefore be formally open to all, while the practical ability to compete depends on access to computational infrastructure, technical expertise, time, and institutional support.

Metric optimisation. Questions of participation also extend beyond access to resources. Participants decide how to respond to the expectations established by a competition, including what aspects of the task to prioritise, how to interpret the evaluation criteria and how much effort to invest (Eisenmann et al. 2023a). As discussed earlier, a single metric provides a clear target around which participants can organise their optimisation, and research has shown that participants actively adapt their development practices and strategies to the metrics and ranking mechanisms through which their solutions are judged (Eisenmann et al. 2023b; Orr and Kang 2024). Yet we know little about how participants understand and experience different evaluation regimes,including whether they regard the criteria as understandable, appropriate, and fair.

Algorithmic monocultures. These issues raise crucial questions not only about who is able to compete, but also about whose knowledge and perspectives are embedded in the resulting models. If the structural conditions systematically privilege particular forms of expertise, resources, or institutional affiliations, the ostensible openness of these competitions may obscure a narrower pool of viable participants. This, in turn, may shape which clinical problems are prioritised and which assumptions become embedded in the algorithms that emerge. Broader work on algorithmic monocultures, has pointed out how repeated reliance on similar models, data, and development practices can reproduce common blind spots across AI systems (Buolamwini and Gebru 2018; Bommasani et al.; Wang, Hertzmann, and Russakovsky 2024). In this sense, questions about who participates in ML competitions are not separate from questions about the systems they produce.

A competition may provide all participants with the same dataset and apply the same metric to every submission, while participants nevertheless differ in their ability to respond to the demands that those arrangements create. Fairness in

competition therefore concerns not only whether algorithms are evaluated appropriately, but also whether participants have meaningful opportunities to participate and compete within the rules of evaluation.

The design of competitions, therefore, is potentially consequential for who is able to contribute to defining what successful AI looks like. Yet we know little about how alternative evaluation designs affect participation. In particular, it remains unclear whether moving from single-metric to multi-metric evaluation changes who participates and persists, how participants navigate the competition, and whether they perceive its conditions as fair.

Conceptual framing

At a high level, we approach machine learning competitions as sociotechnical systems through which algorithmic development and evaluation are organised. Early work examining algorithmic competitions has suggested that their significance extends beyond the technical performance of the systems they produce. In their analysis of the Netflix Prize, Hallinan and Striphas (2016) examined the competition not simply as a mechanism for improving a recommendation algorithm, but as a site in which particular understandings of culture, algorithmic decision-making, and technological expertise were produced. Their analysis drew attention to the broader social and cultural processes through which an apparently technical competition comes to define what is being evaluated, by whom, and for what purposes.

Related work on hackathons and other collaborative technology events has likewise approached these events as sociotechnical systems, focusing on how technologies used for communication and collaboration interact with the social and organisational context of the event. This work highlights how choices made by organisers and participants shape how hackathons are organised and experienced, particularly across different event formats (Mendes et al. 2022; Falk et al. 2024). Such perspectives shift attention from the technical artefacts produced at the end of the event towards the social and technological arrangements through which it is produced.

Work on AI competitions has also begun to make a similar move. Luitse, Blanke, and Poell (2025), for example, conceptualise platform-based medical imaging competitions as infrastructures of power and examine how platforms, organisers, participants, datasets, and computational infrastructures shape the production and evaluation of models. Their analysis shows that competition infrastructures extend across the AI research pipeline, from the definition of research problems and construction of datasets to model production and evaluation. Competitions can therefore be understood as more than technical environments for comparing algorithms; they are organisational arrangements through which algorithmic development is coordinated.

We build on this sociotechnical perspective to broaden our unit of analysis beyond the algorithm and its evaluation. In ML competitions, evaluation does not occur independently of the social and organisational arrangements through which algorithms are produced. Participants interact with particular rules, platforms, computational infrastructures, organisers, evaluation criteria, and ranking mechanisms. These arrangements establish the conditions under which participants develop their solutions and shape how those solutions are subsequently evaluated. We therefore expand our analytical boundary to include the evaluative process and its sociotechnical arrangements. From this perspective, questions of fairness include not only "is the model robust across subgroups?" but also "who gets to participate in building that model in the first place?"

This broadening of the analytical scope further speaks to what Selbst et al. (2019) describe as the framing and formalism traps in fair machine learning. The **framing trap** highlights the risk of defining a system too narrowly, without accounting for the social actors and institutions through which a fairness criterion is realised. The **formalism trap**, on the other hand, concerns the risk of representing social concepts such as fairness through purely mathematical formulations. Selbst et al. (2019) argue that there are multiple legitimate ways of understanding fairness. What counts as fair depends on the social context, the consequences at stake, the procedures considered legitimate, and the perspectives of those involved.

Our study therefore examines ML competitions as sociotechnical systems in which algorithmic development and evaluation are shaped by the interaction of technical, social, and organisational elements. This extends questions of fairness beyond the properties of the algorithms being evaluated to the conditions under which participants develop and compete with them.

Our approach

Within the CHEETAH project, we examine how different competition designs shape who participates and persists. As a first step, we are conducting a scoping review of research on ML competitions across disciplines, examining how competitions are conceptualised and designed and what existing research reveals about participation. The findings will inform the empirical investigation that follows.

We will then focus specifically on evaluation design, comparing single-metric and multi-metric evaluation arrangements. We will examine how these different arrangements shape participation, how participants experience and navigate the competition, and whether these experiences differ across participants with different backgrounds and resources.

We will use a comparative, mixed-methods design grounded in real competition settings that will allow for natural variation in competition design and participant composition.

- Educational machine learning competitions (for example, project courses at our institution)
- Conference-associated competitions (for example, MICCAI)

Comparisons will then be made between traditional single-metric evaluations and novel multi-metric approaches.

Empirical data sources Possible sources for our investigation may include:

1. Competition trace data to analyse entry, persistence, and performance patterns. This would include registrations, submissions, leaderboard/forum activity, with optional voluntary demographics collected with explicit consent and processed in line with GDPR principles.
2. Participant survey administered at different points during the competition to capture expectations and experiences over time.
3. Semi-structured interviews with participants, competition organisers and relevant domain practitioners to capture different perspectives on competition design, participation and fairness. Participant interviews will generate rich accounts of how individuals experience and navigate competitions, including their reasons for participating or withdrawing. Interviews with organises will provide insights into the motivations, constraints, trade-offs underlying evaluation arrangements, while domain practitioners may offer perspectives on the relevance and appropriateness of the the design within their respective domains. Sampling will be purposive, seeking variation in participants' backgrounds and experiences, as well as in stakeholders' roles in competition design and evaluation.
4. Competition artefacts, including rules, metric definitions, documentation, and tutorials, as well as communication materials and judging criteria. These materials help in understanding how visibility and rewards are structured by the competition designers.

Challenges

We recognise several challenges that we may face while conducting this study. First, that participation may be influenced by factors beyond evaluation arrangements including prior experiences, computational resources and individual motivations. Furthermore, competition trace data may reveal patterns of engagement without explaining the decisions underlying them. Therefore, combining these data with participant accounts and organisers' perspectives will help us interpret observed patterns to understand how particular design arrangements are experienced. Accessing participants who withdraw, will however remain a challenge.

Concluding remarks

We bring this emerging research to the workshop to invite discussion of how fairness should be understood when the object of concern is not only the AI systems produced, but also the conditions under which people participate in producing it. We therefore seek feedback from interdisciplinary researchers on how the relationship between evaluation design, participation and fairness in ML competitions might be conceptualised and empirically investigated.

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What Accuracy Cannot Certify: A Failure-Aware Evaluation Framework for Patient-Facing LLM Triage

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Abstract

Patient-facing large language model (LLM) systems are increasingly used to decide how urgently to seek care, yet evaluation still relies heavily on diagnostic or triage accuracy. Drawing on a prospectively registered systematic review of 41 studies (2023–2026), we derive a failure-aware evaluation framework for patient-facing health AI. The synthesis supports three claims beyond the familiar observation that accuracy and safety differ: the clinical significance of error depends on its direction and location along the risk spectrum; measured safety is conditional on the evaluation instrument and interaction design; and the relevant object of safety evaluation is the deployed, versioned system rather than the base model alone. The evidence base also has a deployment-evaluation mismatch: 27 of 41 studies relied on simulated vignettes or authored questions, only 10 incorporated real patient or user data, and only three specifically tested multi-turn dynamics. We therefore propose four evaluation requirements: measure clinically consequential failures directly; stratify them by domain, risk level, and error direction; test realistic language, alternative formulations, and multi-turn interaction; and repeatedly audit the deployed system after material changes. Accuracy remains useful, but it cannot stand in for safety.

Why Accuracy Is an Incomplete Safety Claim

Generative LLMs have moved from research demonstrations into direct-to-consumer health use. A patient may now ask a general-purpose or health-oriented chatbot whether symptoms require emergency care, same-day assessment, or self-management, without a clinician intermediary. Existing reviews have therefore asked a natural question: how accurately do these systems diagnose or assign urgency? Their results suggest moderate self-triage accuracy and, in some settings, triage performance approaching that of healthcare professionals (Kopka, von Kalckreuth, and Feufel 2025; Wang et al. 2026).

For patient-facing deployment, however, the clinically important question is not only how often a model is correct. It is also how it *fails*: whether an error delays emergency care, escalates a benign presentation, omits a crisis safeguard, changes under ordinary patient phrasing, or appears only after several conversational turns. These failure properties are not captured by a single accuracy score.

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This paper makes three claims beyond the familiar distinction between accuracy and safety. First, the clinical significance of error depends on its location and direction along the risk spectrum. Second, measured safety is conditional on the evaluation instrument and interaction design. Third, the relevant object of safety evaluation is the deployed, versioned system—including prompting, retrieval, safeguards, interface, and update state—rather than the base model alone. We derive these claims from a systematic synthesis and translate them into a failure-aware evaluation framework for patient-facing health AI.

The gap between accuracy and safety is also a boundary between communities that evaluate these systems under different commitments. Rather than impose a single definition of safety, we translate the evidence into a framework that technical evaluators can operationalize, clinicians can assess against clinical risk, and governance actors can use to structure oversight.

Evidence Base and Scope

The underlying review was prospectively registered (PROSPERO CRD420261446822; OSF Registries) and followed PRISMA 2020 reporting guidance. Searches of PubMed and Scopus, supplemented by arXiv, medRxiv, targeted web search, and citation chaining, were last run on 9 July 2026. They yielded 475 records before de-duplication and 353 unique records for screening. Three reviewers independently screened titles/abstracts and full texts, extracted data, and appraised study quality. Disagreements were resolved through discussion and consensus. Forty-one studies published from 2023–2026 were included in the narrative synthesis.

The registered scope centered patient-facing triage or self-assessment and required at least one safety-relevant outcome. Four included studies evaluated patient-facing question answering or counselling rather than urgency assignment specifically; these were retained because they directly informed cross-cutting safety constructs such as expert-rated harm, omission, actionability, and failure to recommend care. We use those adjacent studies only for general evaluation claims, not to estimate domain-specific triage failure patterns.

Failure-pattern themes were identified inductively after extraction. Because outcomes were too heterogeneous for meta-analysis, synthesis was narrative. Quality appraisal adapted relevant QUADAS-2 domains and added LLM-specific considerations, including vignette realism and repeated-query reproducibility. Appraisal and study-design mix informed a GRADE-informed certainty assessment of each theme; claims materially dependent on conflicted studies were also sensitivity-checked without them.

1. Safety Has a Clinical Shape That Accuracy Obscures

The clearest evidence comes from studies that measure diagnostic performance and unsafe triage separately. In a retrospective analysis of 40 real emergency-department patients, ChatGPT-3.5 had the joint-highest top-1 diagnostic match rate among the AI tools tested (40%, tied with WebMD), yet also had the highest unsafe-triage rate (41%), significantly higher than the older rule-based Ada symptom checker (14%, $P = .009$). ChatGPT-4.0 had poorer diagnostic accuracy but a lower unsafe rate (22%) and the highest triage agreement with physicians (76%) (Fraser et al. 2023). A model can therefore look better on diagnostic correctness while looking worse on the outcome most relevant to patient harm.

The same separation appears when automated quality metrics are compared with expert judgment. In patient-facing medical question answering, conventional text-similarity measures were essentially uncorrelated with expert-rated harm; a model that performed comparatively well on automated metrics was the worst on expert review (Diekmann et al. 2025). PatientSafeBench similarly found that no evaluated model met its safety bar for patient use, and medical-specialty-tuned LLMs did not reliably outperform general-purpose systems (Kim et al. 2025). These adjacent safety studies reinforce a cross-cutting point: better task performance or greater specialization should not be treated as evidence of safety unless safety itself is measured.

Among direct triage studies, failure has no universal distribution across clinical domains. In physical-emergency triage, a large structured stress-test reported errors concentrated at the two ends of the urgency spectrum, including substantial under-triage of true emergencies (Ramaswamy et al. 2026). The headline magnitude is contested by a partial replication that showed strong effects of response format for some model families, but the disagreement itself strengthens a broader point: measured triage behavior was unstable across evaluation format, model family, and scenario construction (Navarro, Magrabi, and Coiera 2026).

In suicide-risk evaluation, the location of failure can reverse. Across 9,000 responses, one study found reliable behavior at very low and very high risk but poor discrimination among intermediate-risk questions (McBain et al. 2025). Another found that misclassifications of real suicide-support posts clustered between adjacent severity levels, where adjacent categories can separate monitoring from intervention (Patil, Tao, and Gedhu 2025). A generic benchmark that averages across cases can hide these clinically different error shapes. Safety evaluation therefore has to ask not only how many errors occur, but where along the clinical risk spectrum they occur and in which direction.

2. Benchmark Design Can Hide the Failure

The evidence base itself exposes a structural deployment-evaluation mismatch. Twenty-seven of the 41 included studies used simulated vignettes or authored questions, only 10 incorporated real patient or user data, and only three specifically tested multi-turn conversational dynamics. Yet the systems being evaluated are conversational, patient-facing products used with informal language, incomplete information, and changing context.

One red-teaming study reported zero errors across 60 single-turn factual adherence tests but 40–50% error rates after introducing multi-turn adversarial or distress conditions (Hussain et al. 2026). An independent clinical evaluation of suicide-risk conversations found a different multi-turn failure: the tested model progressively withdrew as users disclosed escalating risk and was only 16% as likely to invite continued dialogue at the seventh turn as at the first (Judd et al. 2026). Neither behavior can be characterized by scoring a single, complete vignette-response pair.

Even within single-turn studies, seemingly minor methodological choices changed measured safety. Representative vignettes produced higher measured safety than standard vignettes (Kopka et al. 2024). Holding clinical content constant while changing requester framing produced a 3.3-fold difference in safety-critical under-triage, with patient-oriented phrasing performing worst (Gökmen, Maden, and Zengin 2026). The partial replication of the structured triage stress-test found large forced-choice effects for some model families but not others (Navarro, Magrabi, and Coiera 2026). Together these studies show that vignette representativeness, scoring threshold, response format, prompt framing, and question phrasing are not neutral measurement details; they are potential confounders.

The dominant benchmark can consequently be reassuring precisely because it removes the interaction conditions under which failure emerges. For patient-facing conversational systems, realism is therefore part of measurement validity rather than an optional stress test.

3. Safety Belongs to the Deployed, Versioned System

Evidence for system-architecture and lifecycle effects is more limited than for the first two claims, but it points to an important change in the unit of evaluation. In 1,020 real patient-portal messages, grounding an LLM's emergency-detection decision in a knowledge-graph retrieval system built from a nurse triage protocol book raised sensitivity from 0.73 under prompt-only querying to 0.98 under knowledge-graph global search. Missed emergencies fell from roughly one in four to one in fifty (Liu et al. 2025). The magnitude of this within-study improvement illustrates that system architecture can be an important determinant of safety performance, rather than treating base-model selection as the sole engineering decision.

Observed problem	What accuracy misses	Evaluation requirement
Accuracy–safety gap	A diagnostically strong system can still make more unsafe triage decisions; automated quality scores may not track expert-rated harm.	**Measure consequential failure directly:** report unsafe under-triage, harmful advice, omission, escalation, and other clinically consequential errors alongside conventional accuracy.
Domain-specific failure	Physical-emergency errors can cluster at urgency extremes, while suicide-risk discrimination can fail in the middle.	**Stratify the failure:** report by clinical domain, risk/urgency level, severity, and error direction rather than relying on one aggregate rate.
Design sensitivity	Patient phrasing, vignette construction, output format, and multi-turn interaction can materially change measured safety.	**Test realistic interaction:** use patient language, representative cases, alternative formulations, and multi-turn interaction; report protocol details and score distributions.
Architecture/version dependence	Retrieval, prompting, safeguards, interfaces, and vendor-side updates can alter safety-relevant behavior independently of the nominal base model.	**Audit the deployed system:** identify the tested configuration and version, and trigger re-evaluation after material model, retrieval, prompt, safeguard, or interface changes.

Table 1: A failure-aware evaluation framework derived from observed safety and failure patterns.

Mitigations can also trade one failure for another. Structured prompting raised the proportion of recommendations at least as urgent as the gold standard, but also increased over-triage (Xu, Zhao, and Huang 2025). A safety intervention can therefore make a system more conservative without making it more accurate overall. The correct metric depends on the clinical cost of the error being reduced and the error being introduced.

Lifecycle evidence should also be interpreted cautiously. A longitudinal analysis found large changes in medical safety-disclaimer behavior across model generations (Sharma, Alaa, and Daneshjou 2025); disclaimer frequency is not itself a direct measure of patient harm, but it demonstrates that safety-relevant behavior can be version-dependent. A separate feasibility study proposed version-controlled drift monitoring for patient-facing black-box LLMs because vendor-hosted systems can change without notice to health systems or users relying on them (Harada 2026). A one-time benchmark result therefore describes a versioned system under a particular test configuration; it should not be treated as a durable safety certificate.

A Failure-Aware Evaluation Framework

Table 1 translates the synthesis into four evaluation requirements. The framework does not replace accuracy; it specifies what must be measured alongside it.

Technical, Clinical, and Governance Implications

For technical researchers, the unit of evaluation should be the deployed system: model, prompting, retrieval or other grounding, safety scaffolding, interface, and update process. Robustness testing should vary the interaction conditions that patients actually change—wording, context, turn count, and incomplete information—rather than only the model checkpoint.

For clinicians, the central contribution is to define which errors are clinically consequential and where they sit on the risk spectrum. The same numerical error rate can imply very different harm when it represents delayed emergency care, unnecessary escalation, or imprecision between adjacent mental-health risk categories. Clinical expertise is therefore needed not only to produce gold labels, but to specify the safety construct being measured and the error asymmetries that are unacceptable.

For governance, the lifecycle finding raises a question that the evidence motivates but does not itself legally resolve: *who decides when a changed system requires re-evaluation?* A practical allocation of accountability would make developers responsible for documenting material version and system changes; deployers responsible for deciding when those changes trigger local revalidation; clinicians responsible for defining unacceptable clinical error asymmetries; and evaluators responsible for declaring the model version, system configuration, test conditions, and evaluation date. These are governance implications arising from the evidence, not legal duties established by this review.

Independent replication also matters: three of the 41 included studies had a financial tie to a model they evaluated, so vendor-reported safety claims warrant independent confirmation before deployment decisions.

These perspectives converge on a practical distinction. *Accuracy asks whether the system is correct. Safety asks what happens when it is wrong, uncertain, context-sensitive, or changed after deployment.* A patient-facing system needs evidence for both.

Limitations

This framework inherits the limitations of the underlying review. Thirteen potentially relevant reports could not be obtained; outcomes were too heterogeneous for meta-analysis; no validated appraisal instrument exists specifically for LLM triage safety; and Embase was not searched directly. Evidence strength also differs across claims: domain-specific failure and evaluation-design sensitivity have multiple independent demonstrations, whereas lifecycle effects rest on a smaller, illustrative evidence base. Model-specific numerical results should therefore be read as time-stamped to systems available through mid-2026.

Conclusion

The contribution is not simply that accuracy and safety differ. The synthesis shows *why* accuracy fails as a safety proxy: harm depends on error direction and clinical location; measured safety depends on the evaluation instrument and interaction conditions; and the relevant object of evaluation is the deployed, versioned system. A failure-aware evaluation framework should measure clinically consequential failures directly, stratify them by context and direction, test realistic multi-turn patient use, and continue after material system changes. The relevant question is not simply how often a health LLM is correct, but where, how, and with what consequences it fails.

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